

Effect of weightlifting training on jumping ability, sprinting performance and squat strength: A systematic review and meta-analysis

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Abstract

This systematic review and meta-analysis aimed to assess the effect of using weightlifting movement and their derivatives in training on vertical jump, sprint times, and maximal strength performance. Thirty-four studies were used for meta-analysis with a moderate quality on the PEDro scale. Meta-analysis showed positive effects of weightlifting training, especially when combined with traditional resistance training on countermovement jump performance, sprint times, and one-repetition maximum squat ($ES = 0.41$, $ES = -0.44$, and $ES = 0.81$, respectively). In conclusion, results revealed the usefulness of weightlifting combined with traditional resistance training in improving sprint, countermovement jump and maximal strength performance.

Keywords

Acceleration, countermovement jump, resistance training

Introduction

Weightlifting movements are ballistic resistance exercises that involve great acceleration of the segments through the entire movement resulting in mobilization of the barbell at high velocities.¹ There are two kinds of exercises, the snatch (SN) and the clean & jerk (C&J), which are performed in the sport of weightlifting.² Outside of weightlifting competitions, weightlifting derivatives such as hang snatch, clean pull, and push jerk exercises are commonly used during strength and power training programs.^{3–8}

The use of weightlifting is popular in both fitness and sports fields. In particular, weightlifting exercises have been suggested as a useful tool to improve jump height and muscular strength⁹ due to the biomechanical, neural, and muscular characteristics of these movements. According to the literature, several reasons could be argued in favor of weightlifting exercises. From a neuromuscular perspective, compared with traditional exercises such as the squat or deadlift, weightlifting exercises show decreases in muscles' coactivation.¹ These differences in the neuromuscular pattern are linked to improvements in muscle-tendon unit stiffness,^{10,11} resulting in an optimization of the stretch-shortening cycle.¹² Another important advantage of weightlifting exercises is not only the great number of muscle groups recruited during these movements¹³ but also the

substantial levels of muscle activation achieved,¹⁴ similar to isolated traditional exercise.¹⁵ Compared to traditional resistance exercises, weightlifting movements show greater velocity and power values.¹⁶ This can be potentially explained by the ballistic component and the nature of the weightlifting exercises, since they allow to apply force during a greater range of movement than traditional exercises.^{17–20} Besides, the second pull of weightlifting is characterized by an explosive triple extension action of the ankle, knee, and hip joints,²¹ which is similar to main sports actions such as a vertical jump^{22–26} and a sprint.²⁶ Thus, the inclusion of weightlifting movements within strength training programs is meaningful, due to the potential transfer to many sports actions' performance.^{27–29}

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Despite the widely reported usefulness of weightlifting movements in improving sprint and jumping performance,^{3,30,31} their superiority to other resistance training methods is controversial.³² However, from a practical point of view, most athletes' training programs use a combination of traditional and weightlifting exercises. Thus, the present review and meta-analysis aimed to assess the effectiveness of isolated weightlifting training, isolated traditional resistance training, and a combination of them in improving one repetition maximum (1RM), jumping and sprinting performance.

Methods

A systematic review of the literature and meta-analysis were conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.³³

Data search and sources

Potential studies were identified via a comprehensive strategy. The electronic database search was carried out in PubMed, Scopus and Google Scholar database (restricted to the first 1000 citations) from the inception up to 12 December 2020. The search strategy included the terms "Weightlifting", "Snatch", "Power clean", "Hang clean", "Clean and jerk", "Split jerk", "Push jerk", "High pull", "1RM", "Jump", and "Sprint". References from the original studies included were searched for further relevant investigations. Besides, the authors of the included articles were contacted to identify possible additional published and unpublished studies.

Study selection

Eligibility criteria were established according to PIO (participants, intervention, and outcomes) guidelines. We selected studies performed with healthy male or female participants regardless of age. All studies should have weightlifting exercise included in the experimental group or compared weightlifting with other training methodologies. In this sense, we sorted the intervention groups as weightlifting in isolation (WL group), traditional training (TD group) and combined training (CB group). The classification was made according to the training volume (i.e. sets $\times$ repetitions) of lower-limb exercises because output analysis in this meta-analysis refers to lower-limb performance. Interventions were allocated in the WL group if athletes performed weightlifting (SN or C&J) exercise or its derivatives from catching (exercises that include catch phase) and pulling (exercises that exclude catch phase) in isolation or training volume of weightlifting exercises were more than 50% of total training volume. If weightlifting exercises were performed with other strength exercises and

weightlifting training volume was less than 50% of total training volume, they were included in the CB group. Strength training that did not meet the criteria for being included in the CB or WL group, was allocated to the TD group. Each selected study had to assess 1RM squat, sprinting -up to 40-yards, and/or jump performance and agree with a single and multi-intervention study. Only articles published in English were selected. For studies with more than one article based on the same sample, inclusion was limited to just one of them.

Three reviewers (R.S., J.L.H., A.G.) independently analyzed the titles and abstracts of the articles retrieved from the literature search and reviewed the full text of the selected articles. Any disagreement between the reviewers regarding study inclusion was resolved by a fourth investigator (A.M.).

Methodological quality

Reviewers classified articles according to PEDro scale for reporting on the methodological quality.³⁴

Computation of effect sizes and statistical analysis

The literature search revealed studies that included a control group (CG) and studies without a CG. To not discard studies without a CG, we included all of them and, as a consequence, our analysis unit was the group, not the study. Thus, for each group, the standard mean difference (SMD) was used as the effect size (ES) index. SMD was defined as the difference between pretest and post-test means divided by the standard deviation at baseline, then corrected by the factor for small samples ($df_{E,C}$).³⁵ Although between-group ES exhibits better internal validity than within-group ones, using one-group pretest-post-test ES is recommended in the meta-analytic arena when there are many studies without a CG.³⁶ To perform statistical analysis, a random-effects model was applied using the restricted maximum likelihood (REML) estimator for calculating within-study variance estimated, which has been proved to be reasonably robust even in the absence of normal data distribution.³⁷ A first analysis consisted of calculating the mean ES with its 95% confidence interval (CI), the heterogeneity test, Chi^2 , and the I^2 index.^{38,39} Statistical analyses were considered significant at $p < .05$. The practical relevance of each ES was interpreted with the Cohen's criterion (trivial: < 0.2 ; small: $0.2-0.49$; moderate: $0.5-0.79$; large: ≥ 0.8).⁴⁰ The heterogeneity was interpreted depending on I^2 magnitude as no heterogeneity ($< 25\%$), low ($25-49.9\%$), moderate ($50-74.9\%$), and large heterogeneity ($> 75\%$).^{38,41}

In case of substantial heterogeneity ($Chi^2 < 0.05$ and/or $I^2 > 50\%$), random-effects meta-regression for continuous moderator variables (i.e. participant age, total training session, and publication year) and subgroup analysis for

categorical moderator variables (i.e., training frequency [≤ 2 , 3, and ≥ 4 days a week], length [≤ 8 , 8–12, and >12 weeks], and performance level) were carried out to explain the heterogeneity. The performance level was sorted by 1RM squat lift according to the strengthlevel.com database. Nevertheless, studies that did not provide strength values were sorted by comparing height CMJ to Hoffmann et al.⁴² In both cases, the cut-off value was determined to percentile 5, 20, 50, 80, and 95 for beginner, novice, intermediate, advanced, and elite category, respectively.

To investigate the robustness of our results, sensitivity analyses were carried out. We assessed the impact of including or excluding outliers. To assess outliers, we used the outlier-labelling rule.⁴³ Besides, publication bias analyses were carried out through the Egger test⁴⁴ and the nonparametric trim-and-fill method.⁴⁵

Statistical analyses were performed using RStudio (version 4.0.2) with the “meta” package (version 4.14), “dmetar” package (version 0.0.9000), and “metafor” package (version 2.4).

Results

Literature search

The literature search process resulted in a total of 2350 articles. Initially, all duplicate articles were removed (122 articles dismissed). A total of 2228 remained, and a further 2157 were removed after screening using the title and abstract. The full text of the remaining 71 articles were reviewed for more detailed evaluation and resulted in the exclusion of 37 articles. Based on the eligibility criteria, 34 articles were included in the final analysis (Figure 1).

Methodological quality

All studies included in the meta-analysis were classified by the PEDro scale and had a range of 4–7 points. Thus, they were considered to be of moderate quality. The 34 studies included along with the result of the PEDro scale are described in Table 1.

Outcomes

Pooled analysis showed a statistically significant improvement after training in all analyzed variables ($p < .05$), and the ES magnitudes showed a small to moderate change for SJ (number of analysis units [k] = 34; n = 372; SMD_+ = 0.31 [95% CI = 0.22, 0.40]; Figure 2), CMJ (k = 43; n = 536; SMD_+ = 0.33 [95% CI = 0.21, 0.46]; Figure 3), sprint (k = 23; n = 304; SMD_+ = -0.38 [95% CI = -0.55, -0.21]; Figure 4), and 1RM squat (k = 28; n = 345; SMD_+ = 0.58 [95% CI = 0.38, 0.79]; Figure 5).

Subgroup analyses showed no statistically significant differences between training methods (i.e. CB, WL, and TD) for

SJ (Q_M = 0.29; p = 0.866), CMJ (Q_M = 3.54, p = 0.171) or sprint (Q_M = 0.81, p = 0.666), but for 1RM (Q_M = 10.17, p = 0.006). However, heterogeneity analyses of the overall SMD reached statistical significance with moderate inconsistency for CMJ (I^2 = 51%, p < 0.01), sprint (I^2 = 56%, p < 0.01), and 1RM (I^2 = 60%, p < 0.01), while no inconsistency was found for SJ (I^2 = 0%, p < 0.630). Therefore, moderator analyses were performed for CMJ, sprint, and 1RM.

Analysis of moderator variables on the ESs of CMJ, sprint, and 1RM squat

Meta-regression analysis did not show results influenced by continuous moderators' variables described before. However, several categorical variables showed an influence on the training-induced effect on CMJ (Table 2), sprint (Table 3), and 1RM (Table 4) results. Regarding moderator effect in CMJ, the higher frequency of training showed a higher ES (SMD_+ = 0.51) than the lower frequency (SMD_+ = 0.21). For the sprint, the performance level influenced the result, since the inexperienced participants (SMD_+ = -0.71) achieved better results than the experienced (SMD_+ = -0.38). On the other hand, for 1RM, the length of the intervention was positive for 1RM squat (SMD_+ = 0.94) regardless of increasing frequency (SMD_+ = 0.53).

Sensitivity analysis

Outliers were found for the SJ, CMJ, sprint, and 1RM sensitivity analysis. Thus, for SJ with a lower critical value (SMD = 0.22) and an upper critical value (SMD = 0.40), the outlier was Sofiene et al.⁶⁸ (SMD = 0.97). For CMJ, with a lower critical value (SD = 0.21) and an upper critical value (SMD = 0.46), the outliers were Chaouachi et al.⁴ (SMD = 1.39), Ince⁵⁵ for the split group (SMD = -0.33) James et al.⁵⁸ for weaker group (SMD = 3.56), and Loturco et al.⁵⁹ (SMD = -0.89). For sprint, with a lower critical value (SMD = -0.55) and an upper critical value (SMD = -0.21), the outlier was Moore et al.⁶² (SMD = -2.73). For 1RM squat, with a lower critical value (SMD = 0.38) and an upper critical value (SMD = 0.79), the outliers were Hawkins et al.⁵³ (SMD = 2.45), Moore et al.⁶² (SMD = 13.24), and Slovak et al.⁶⁷ (SMD = 3.46).

The analysis was carried out after removing the outlier. The conclusion was similar to before removing the outliers in all variables (Table 5). In contrast, the heterogeneity showed a stronger decrease for CMJ (from I^2 = 51% to I^2 = 26%, p = 0.07) and 1RM (from 60% to 43%, p < 0.05).

Publication bias

The Egger tests were statistically significant for SJ ($t[33]$ = 2.06, p = 0.047), CMJ ($t[42]$ = 2.38, p = 0.022), and 1RM squat ($t[27]$ = 5.76, p < 0.001). Therefore, the trim-and-fill

Table 1. Characteristics of studies.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Arabatzi and Kellis ¹	6	Male students ($n = 26$; 21.5 ± 12.5 y)	WL ($n = 9$)	Power clean; snatch; clean & jerk; high pull; half-squat	4–6 series $\times$ 6 rep (weeks 1–4) 4 series @ 4RM (weeks 5–8)	3 d·wk ⁻¹	8 wk	Hand on hips vertical jump on a force platform (Type 9281C, Kistler Instruments, Winterthur, Switzerland)	OW = TD > C
			TD ($n = 9$)	Leg press; leg curl; leg extension; bench press; half-squat	4–6 series $\times$ 6 rep (weeks 1–4) 4 series @ 4RM (weeks 5–8)				
Ayers et al. ⁴⁶	6	NCAA female athlete ($n = 23$; 21.1 ± 1 y)	C ($n = 8$) WL HC ($n = 11$)	Their standard sport activities They included hang clean in their standard training.	NA 5 series $\times$ 3 rep @ 80–85% 1RM	2 d·wk ⁻¹	6 wk	1RM back squat. 40-yard sprint (hand-held stopwatch) Vertical jump on Vertec (Sports Imports, Columbus, Ohio)	HC = HS
Barfield and Anderson ⁴⁷	5	Male students ($n = 50$; 21 ± 2.5 y)	WL HS ($n = 12$) CB ($n = 25$)	They included hang snatch on their standard training. Include varied gymnastic and Olympic lifts	CrossFit™ referenced training	2 d·wk ⁻¹	14 wk	1.5-mile run. Vertical jump. Waist-hip ratio. Body mass index YMCA sit-and-reach	OW = TD
			TD ($n = 25$)	Multi-joint exercises lie: Bench press; Deadlift; Squat	3 series $\times$ 8–12RM 3–5 series $\times$ 2–3 RM 3–5 series $\times$ 2–3 rep @ low-moderate intensity				
Barr ⁴⁸	4	University Canadian football players ($n = 31$; 20.8 ± 1.9 y)	WL ($n = 31$)	Variations of snatch; clean & jerk; back squat; pressing exercises	Weekly repetition: 10–15 @ 60–90% 1RM 10–12 @ +90% 1RM	3 d·wk ⁻¹	16 wk	1RM front squat. 1RM press. 1RM clean & jerk. Countermovement jump (just jump, probotics, huntsville) Body mass index.	Pre-WL < Post-OW
Channell and Barfield ⁴⁹	5	Male students ($n = 27$)	WL ($n = 11$)	Including power clean, hang clean, clean pulls and snatch on exercise progression	3–5 series $\times$ 10–3 rep @ 75–95% 1RM	3 d·wk ⁻¹	4 + 8 wk	1RM back squat. Vertical jump on Vertec (Sports Imports, Columbus, Ohio.)	OW > PT > C
			TD ($n = 10$)	Including squat, overhead squat, deadlift and leg press on exercise progression					
Chaouachi et al. ⁴	6	Children ($n = 64$; 10–12 y)	C ($n = 6$) WL ($n = 17$)	Their standard sport activities Clean, snatch, shoulder push press, kettlebell/cross-body pull	NA 1–3 series $\times$ 12–18 rep @ maximal weight they manage to lift with correct technique	3 d·wk ⁻¹	12 wk	Body mass index. CMJ. (Quattro-Jump; Kisler, Winterthur, Switzerland) Horizontal jump. Balance.	OW > PL > TD > C

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
								5- and 20-m (Brower Timing Systems, Salt Lake City, UT, USA) Isokinetic force at 60 and 300°s ⁻¹ .	
Ciacci and Bartolomei ⁵⁰	6	Male basketball (n = 58; 18.3 ± 1.8 y)	PL (n = 17) TD (n = 17) C (n = 13) CB senior (n = 10) CB U19 (n = 10) CB U17 (n = 9) TD senior (n = 12) TD U19 (n = 8) TD U17 (n = 9)	Maximum CMJs, drop jumps, drop from a low platform, Medicine ball throws. Squat, lunges, chest press, shoulder flyes or presses Their standard sport activities Hang clean; jump rope Half squat; speed ladder	NA 4 series × 8 rep @ 50% IRM +CMJS 200 foot contacts/session 4 series × 8 rep @ 50% IRM +CMJS 200 foot contacts/session	2 d ·wk ⁻¹	16 wk	Bosco test	OW1 = TD1 OW2 = TD2 OW3 < TD3
Comfort et al. ⁸	7	Youth soccer player and Collegiate athletes (n = 34; 19.5 ± 2.3 y; 26 male and 8 female)	WL catch (n = 16)	Power clean, mid-thigh power clean, back squat, push press, Nordic lowers	3 series × 5 rep @ 67.5–82.5% IRM (weeks 1–4) 3 series × 3 rep @ 75–90% IRM (weeks 6–9)	2 d ·wk ⁻¹	12 wk	Hand on waist vertical jump testing on a force platform (type 9286AA, Kistler Instruments Inc., Amherst, NY, USA) Maximal isometric mid-thigh pulls. IRM Power Clean.	Catch = Pull
Gharib ⁵¹	5	Federated weightlifters (n = 12; 19.5 ± 3.05)	WL pull (n = 18) WL (n = 6)	Clean pull, mid-thigh pull, back squat, Push press, Nordic Lowers Snatch power, back drop in large grip, snatch squat, deadlift large grip, clean power, front squat, back squat, deadlift narrow grip, power jerk, back jerk, jerk from split position, quarter squat, snatch pull and snatch.	2–3 series × 1–4 rep @ 75–80% IRM	3 d ·wk ⁻¹	12 wk	Strength tests (front squat, back squat and good morning) power test (clean power and snatch power) balancing test (snatch balance and deadlift)	OW > C
González-Badillo et al. ⁵²	6	Junior male lifter (n = 51)	C (n = 6) WL 1 (n = 16) WL 2 (n = 17) WL 3 (n = 18)	Their standard training Snatch, clean & jerk, pulls, squat	NA 3–7 series × 2–6 rep @ 60–80% IRM 1–3 series × 1–3 rep @ 90–100%	4–5 d ·wk ⁻¹	8 wk	IRM snatch IRM clean & jerk IRM squat	OW 1 = WL 2 = WL 3
Hawkins et al. ⁵³	6	College-aged male (n = 29; 21.5 ± 12.5 y)	TD (n = 10)	Squats, lunges, bent-over row, pullover, upright row, deadlift,	3 series × 10–6 rep (weeks 1–4)	3 d ·wk ⁻¹	4 + 4 wk	CMJ and SJ on a force plate (Fitness Technology, Skye,	OW > PL OW > TD

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Helland et al. ³²	6	Young athletes (n = 39; 20 ± 3 y)		stiff-leg deadlifts, step-up, walking lunges, split squats, stiff-leg deadlifts, 45-degree lunges, standing 1-arm row, standing dumbbell press.	3 series × 6–4 rep (weeks 5–8)			South Australia) 1RM Squat.	
			PL (n = 10)	Jumps in place, squat jumps a-skips, single-leg hops, double-leg hops, double-leg hops, jump and reach, double-leg tuck jump, depth jump bounding.	3 series × 10–13 rep (weeks 1–4) 3 series × 15–5 rep (weeks 5–8)				
			WL (n = 9)	High pull Hang clean, Power push (front), Snatch balance, Front squats, Split jerks, Back squat.	3 series × 8–6 rep (weeks 1–4) 3 series × 6–2 rep (weeks 5–8)				
			WL (n = 13)	Clean with front squat; hang clean; snatch; power jerk; power clean; hang snatch;	3–5 series × 3–5 rep @ 5–3RM	2–3 d ·wk ⁻¹	8 wk	CMJ SJ DJ 30-m sprint 1RM Squat	IK > TD > OW
Hoffman et al. ³	6	NCAA Division III football team male (n = 20; 19.1 ± 1.3 y)	IK (n = 13)	Squat 0.4 m/s; Single leg squat 0.4 m/s; CMJ; Single leg CMJ	1–4 series × 5–10 rep @ 40–60% 1RM +120–140% ecc				
			TD (n = 13)	Squat; single leg squat; CMJ; single leg CMJ	2–6 series × 5–3 rep @ 40–80% 1RM/5–3RM				
			CB (n = 10)	Bench press, Incline bench press, Incline flys, squats, deadlift, leg extension, snatch pulls (knee), snatch pulls (floor), dumbbell pulls, push jerk, clean (floor), clean pull (knee), power shrugs, overhead squats, etc. Between other exercises as sprinting	4–2 series @ 10RM (week 1–5) 4–5 series @ 6–8, 5RM (weeks 6–10) 4–5 series @ 6–8, 5RM (weeks 11–15)	4 d·wk ⁻¹	15 wk	1RM Squat 1RM Bench press CMJ (Celesco model PT 9510, Canoga Park, CA) 40-yard sprint (hand-held stopwatch) T-drill (hand-held stopwatch)	OW > PL
			TD (n = 10)	Bench press, incline bench press, incline flys, squats, deadlift, leg extension, stiff leg deadlift, leg curl, incline dumbbell bench press, upright row, front raise, etc. Between other exercises as sprinting	4–2 series @ 10 RM (week 1–5) 3–4 series @ 6–8RM (week 6–10) 3–5 series @ 6–8RM (week 11–15)				
			WL female (n = 3) WL male (n = 4)	Squat, front squat, standing press, snatch grip, clean grip, clean grip shoulder shrugs, clean grip mid-thigh pull, stiff legged deadlift, snatch grip shoulder shrugs, undulating, snatch, snatch grip stiff-legged deadlift	1 series × 4 rep @ 60–95%	2 d·wk ⁻¹	20 wk	Isometric Mid-Thigh Clean Pull (Rice Lake Weighing Systems, Rice Lake, WI, USA) Static Vertical Jumps	Pre < Post
Ince ⁵⁵	6	Collegiate female			3–5 series × 5 rep @	2 d·wk ⁻¹	6 wk	Spike jump	

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Ince ⁵⁶	6	volleyball players (n = 33; 15.38 ± 0.98 y)	WL Split (n = 11)	Split hang power snatch, split hang power clean, split jerk	70–90% 1RM rest: 2 min.			CMJ (Optojump Next. Microgate, Bolzano) Change of direction Leg stiffness 5-m sprint 20-m sprint (Photocell. Microgate, Bolzano)	Split = Squat > C
			WL Squat (n = 11)	Squat hang power snatch, squat hang power clean, power jerk					
			C (n = 11) WL (n = 11)	Clean pull and snatch pull, block snatch pull and block clean pull, hang high pull, jump shrug	3 series × 3–5 rep @ 30–100%	2 d·wk ⁻¹	8 wk	1RM Snatch 1RM Clean & Jerk 1RM Front squat CMJ (Optojump Next®, Microgate, Bolzano, Italy) Peak barbell velocity (Tendo Weightlifting Analyzer; Tendo Sports, Trencin, Slovak Republic) Back strength (Fabrication Enterprises, Inc., New York, USA)	OW = PL > WL (control)
Ince ¹²	6	Collegiate volleyball players female (n = 34; 15.43 ± 1.91 y)	PL (n = 12)	Hurdle hops, Box jump, Depth jump, Medicine ball back throws	2–4 series × 5–20 rep				
			WL (Control) (n = 11) WL (n = 17)	Snatch, Clean & Jerk, Squat, Power Snatch and Clean & Jerk Hang split snatch, Hang split clean, Split jerk were included in their standard training.	6–8 series × 1 rep @80–95% 2–4 series × 5 rep @ 70–85% 1RM Increasing load every week	2 d·wk ⁻¹	6 wk	Spike jump (Optojump) CMJ (Optojump) 5- and 20 m sprint (Microgate, Bolzano, Italy) Change of direction (Microgate, Bolzano, Italy) Leg stiffness	OW > C
James et al. ⁵⁷	4	Males (n = 16)	C (n = 17) CB stronger (n = 8) CB weaker (n = 8)	Their standard training and activities Power clean, jump squat, hang power clean, snatch grip pull, depth jump, plyometric split squat (rebound)	Week 1–5 4–5 series × 5 rep @ 40–70% 1RM Week 6–10 4–5 series × 4–3 rep @ unload-85% 1RM	3 d·wk ⁻¹	10 wk	1RM Power Clean CMJ (Bertec Corporation, USA)	Pre < Post
James et al. ⁵⁸	4	Males (n = 16)	CB stronger (n = 8)	Power clean, jump squat, hang	5 series × 5 rep @ 40–	3 d·wk ⁻¹	10 wk	1RM Squat	

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Loturco et al. ⁵⁹	5	Male soccer player (n = 17; 18.4 ± 1.2 y)	= 8) CB weaker (n = 8)	power clean, snatch grip pull, depth jump.	70% 1RM 5 series × 4 rep @70–85% 1RM 3–5 series × 3–4 rep	2 d·wk ⁻¹	6 wk	Jump Squat Isometric squat Muscle activation	OW weaker > WL stronger
			WL (n = 8)	Push-press	Session 1–4 6series × 8 rep @ optimum power load Session 5–8 6 series × 6 rep @ 1.05 optimum power load Session 9–12 6 series × 4 rep @ 1.1 optimum power load			Squat jump CMJ 5-m sprint 10-m sprint 20-m sprint 30-m sprint Change of direction Speed tests	TD > OW
Maulit ⁶⁰	6	Recreationally resistance-trained men (n = 31; 23.13 ± 2.38 y)	WL EDLG (n = 16)	Explosive deadlift (NSCA)	Linear periodization training model: Beginning 4 series × 5 rep @ 30% 1RM (week 1–2) Ending 6 series × 4 rep @ 40% 1RM (week 3–4)	2 d·wk ⁻¹	4 wk	Vertical jump (EPIC, JS; Lincoln, NE, USA) 1RM deadlift Maximal isometric Mid-Thigh Pull	Pre < Post
Miller ⁶¹	4	Male football player and female soccer player (n = 21; 19.85 ± 0.88 y)	CB female (n = 12) CB female (n = 9)	Clean, snatch pull, hang snatch split jerk, power jerk, clean grip RDL, back squat, snatch grip lunges, incline press, bar rows, single-arm dumbbell bench, pull-ups/ pull-downs, single leg RDL, Front squat, Side lunge, bench press (...)	3–6 series × 12–5 rep @ 70–95% 1RM	4 d·wk ⁻¹	6 wk	1RM Clean 1RM incline press 1RM squat Body composition	Pre < Post
Moore et al. ⁶²	5	Collegiate athletes (n = 15; 20.2 ± 0.2 y; 10 females, 5 males)	CB (n = 8)	Hang clean, Romanian straight leg deadlift +Resistance training supersets	3 series × 6 rep 3 series × 6 rep @ failure	3 d·wk ⁻¹	12 wk	CMJ on Vertec (Sports Imports, Inc., Columbus, OH) 4RM Squat 25 m sprint (hand-held	OW = PL

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
								stopwatch) Figure 8 drill on a 5-dot mat	
Oranchuk ⁶³	6	Female and male golfers (n = 12; 20.3 ± 1.5 y)	PL (n = 7)	Hops, over & back, side-to-side, one-legged moving, bounds, tuck jumps, split squat jump, bound hop back, box jump, stadium hops, alternate bound, one-leg zig-sag +Resistance training superset	2–3 series x 15–30 rep (week 1–4) 2 series x 15–30 rep (week 5–6) 1–3 series x 15–30 rep (week 7–8) 1–3 series x 10–16 rep (week 9–10) 1–3 series x 10–30 rep (week 11) 3 series x 6 rep @ failure	3 d·wk ⁻¹	8 wk	Club head speed CMJ (Just Jump, Probotics Inc, Huntsville, AL, USA) IRM squat IRM power clean IRM deadlift	OW > C
			C (n = 6)	Bodyweight single-leg squat, dumbbell single-leg RDL, leg curl, plate sit-up, medicine-ball twist, pull-up, dumbbell one-arm bench, band pull-apart, face-pulls, medicine ball golf swing, medicine ball walking lunge, cable internal rotation, cable wood-chop, medicine ball sit-up, one-leg back extension	Week 1–4 3–5 series x 10–4 rep @ 85–90%/60–120 s Week 5–8 3–5 series x 8–3 rep @ 92.5–95%/60–120 s 3–4 series x 12–4 rep/30–45s				
Oranchuk ⁶⁴	7	NCAA swimmers (females n = 10; males n = 8; 19.6 ± 2.7 y)	CB (n = 9)	Hang high-pull, back squat, dumbbell rdl, alternating dumbbell bench, one-arm cable row, bench press, pull-ups, dumbbell single-leg deadlift, lateral lunges	Week 1–4 3–4 series x 12–4 rep @ 10–75% / 60–120 s Week 4–8 3–6 series x 10–3 rep @ 20–85%/60–180 s Week 9–10 2–5 series x 10–2 rep @ 25–90% / 60–180 s	2 d·wk ⁻¹	10 wk	SJ CMJ Isometric mid-thigh pull	OW = TD
			TD (n = 9)	Trap-bar jump squat, Back squat, Dumbbell RDL, Alternating dumbbell bench, One-arm cable					

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Otto ³⁰	6	Healthy men (n = 30; 22.82 ± 1.91 y)	WL (n = 13)	row, Bench press, Pull-ups, Dumbbell single-leg deadlift, Lateral lunges High pull, Power clean, Back Squat	3–4 series × 6–4RM (weeks 1–3) 4–6 series × 6–4RM (week 4–6)	2 d · wk ⁻¹	6 wk	IRM Back Squat IRM Power Clean Vertical jump (EPIC Athletic Performance, Inc., Colorado Springs, CO, USA)	OW > KT
			KT (n = 17)	Kettlebell swing, Accelerated swings, goblet squat	3–4 series × 6–4 (weeks 1–3) 4–6 series × 6–4 (weeks 4–6)				
Pichardo et al. ⁶⁵	6	Male students (n = 59; 12–14 y)	CB (n = 31)	Clean derivatives (above the knee), snatch derivatives (above knee), squat, lunge, upper body horizontal push and pull, brace, rotate, hinge, reactive horizontal and vertical jump, lineal speed.	Weeks 1–6 1–3 series × 8–20 rep Weeks 7–14 1–3 series × 8–12 rep Week 15–22 1–4 series × 2–6 rep Weeks 23–28 1–4 series × 2–5 rep	2 d · wk ⁻¹	28 wk	Resistance training skills battery quotient Isometric mid-thigh pull (Pasco, CA, USA) CMJ (GymAware; Kinetic Performance Technology, Canberra, Australia) Horizontal jump 10-m sprint 20-m sprint 30-m sprint Seated medicine ball throw	OW = TD
			TD (n = 28)	Deadlift + vertical plyometric jump, Sumo deadlift + horizontal plyometric jump, squat, lunge, upper body horizontal push and pull, brace, rotate, hinge, reactive horizontal and vertical jump, lineal speed.					
Roberts ⁶⁶	6	High school football players male (n = 40; 14–18 y)	CB (n = 20)	Hang clean, hang power clean, high pull. (Only 2 d · wk ⁻¹ Olympic lifting exercise) Bench press, squat and other auxiliary lift	3 series × 5 rep @ 60–95% 1RM (increasing 5% 1RM per week)	4 d · wk ⁻¹	8 wk	IRM Squat IRM Power clean Vertical jump (Probotics Just Jump system) 9.14 m sprint (Polaris Sports Timing System)	OW > TD
			TD (n = 20)	Deadlift, Bench press, squat and other auxiliary lift					
Slovak et al. ⁶⁷	4	Young female handball athletes (n = 10; 17 ± 1.3 y)	CB (n = 10)	Squat, Snatch pull, Full snatch, Second round in the throwing gesture, Bench press	3 series × 5–10 rep @ load: 22–46	NA	8 wk	Abalakov jump CMJ (Globus Brasil, RJ, Brazil) IRM squat 10-m sprint	OW > TD

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Sofiene et al. ⁶⁸	6	Elite table tennis player (n = 30; 18.05 ± 1.04 y)	TD (n = 10)	Squat, Earth lifting, Stiff, Bench press, Bent-over row	3–4 series × 3–10 rep @ load: 30–58	3 d·wk ⁻¹	6 wk	20-m sprint 30-m sprint	OW > C
			WL (n = 15)	Power snatch, power clean Squat, Push press, Jerk	2–4 series × 4–8 rep @ 70–85% 1RM			5-m sprint SJ CMJ Strength hand grip Throwing medicine ball 1RM Bench press 1RM squat	
Suchomel et al. ⁶⁹	6	Male (n = 27; 22.43 ± 2.52 y)	C (n = 15)	Their standard training and activities	3–5 series × 2–10 rep @ 50–82.5% 1RM Power clean	3 d·wk ⁻¹	10 wk	Isometric mid-thigh pull 1RM Power clean 10-m sprint 20-m sprint 30-m sprint 505 change of direction	OW Overload > WL Catch = WL Pull
			CB Catch (n = 9)	Power clean, Midhigh power clean, Countermovement power clean, hang power clean					
			CB Pull (n = 9)	Clean pull, Midhigh pull, Countermovement shrug, hang high pull, jump shrug					
			CB overload (n = 9)	Clean pull, midhigh pull, clean pull, countermovement shrug, hang high pull, jump shrug					
Suchomel et al. ⁷	6	Male (n = 25; 22.64 ± 2.36 y)	CB catch (n = 8)	Power clean, midhigh power clean, Countermovement power clean, hang power clean	3–5 series × 2–10 rep @ 50–82.5% 1RM Power clean	3 d·wk ⁻¹	10 wk	SJ (PASPORT, PASCO Scientific, CA, USA) CMJ (PASPORT, PASCO Scientific, CA, USA)	OW Overload > WL Catch = WL Pull
			CB pull (n = 9)	Clean pull, midhigh pull, countermovement shrug, hang high pull, jump shrug	3–5 series × 2–10 rep @ 50–82.5% 1RM Power clean				
			CB overload (n = 8)	Clean pull, midhigh pull, clean pull, countermovement shrug, hang high pull, jump shrug	3–5 series × 2–10 rep @ 30–110% 1RM Power clean				
			WL (n = 13)	Hang power clean, power snatch, half squat,	4 series × 4 rep @ 70% 1RM; 4 series × 6RM (week 1–3) 6 series × 4 rep @ 70% 1RM; 4 series × 6RM (weeks 1–3)				
Teo et al. ⁷⁰	7	Recreationally active male (n = 26; 24.2 ± 1.11 y)	PL (n = 13)	Weighted double, leg jumps, 45 cm depth jumps, half squat	4 series × 4 rep @ 30% 1RM; 8 series 10 rep; 4 series × 6RM (week 1–3) 6 series × 4 rep @ 30% 1RM; 8 series 12	3 d·wk ⁻¹	6 wk	CMJ SJ DJ 5- and 20 m sprint 505 agility test	OW = PL

(continued)

Table 1.continued.

Study	PEDro's score	Participants	Experimental group	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Tricoli et al. ³¹	5	Male college physical education students (n = 22; 22.0 ± 1.5)	WL (n = 7)	High pull, Power clean, Clean and Jerk	rep; 4 series × 6RM (week 1–3) 3–6 series × 6–4RM (weeks 1–4) 4–6 series × 6–4RM (weeks 5–8)	3 d·wk ⁻¹	8 wk	SJ and CMJ (Jump Test Pro device) 10- and 30 m sprint (photocells) Agility test (photocells) IRM Clean and jerk IRM Half Squat	OW > PL > C
			PL (n = 8)	Double-leg hurdle hops, Alternated single-leg hurdle hops, single-leg hurdle hops, 40-cm drop jumps	4 series × 4 rep (weeks 1–4) 6 series × 4 rep (week -8)				
			C (n = 7)	No training					

y: Years; rep: Repetition; d: Days; wk: Week; CMJ: Countermovement Jump; SJ: Squat Jump; DJ: Drop Jump; NA: Not an available; WL: Weightlifting and its derived; TD: Traditional Weightlifting; C: Control; PL: Plyometric Training; IK: Isokinetic; HC: Hang Clean; HS: Hang Snatch; Catch: Power Clean with Catch phase; Pull: Power Clean without Catch phase; KBG: Kettlebell Training; EDLG: Explosive Deadlift Training; RDL: Romanian deadlift; RPE10: 0–10 rate of perceived exertion scale.

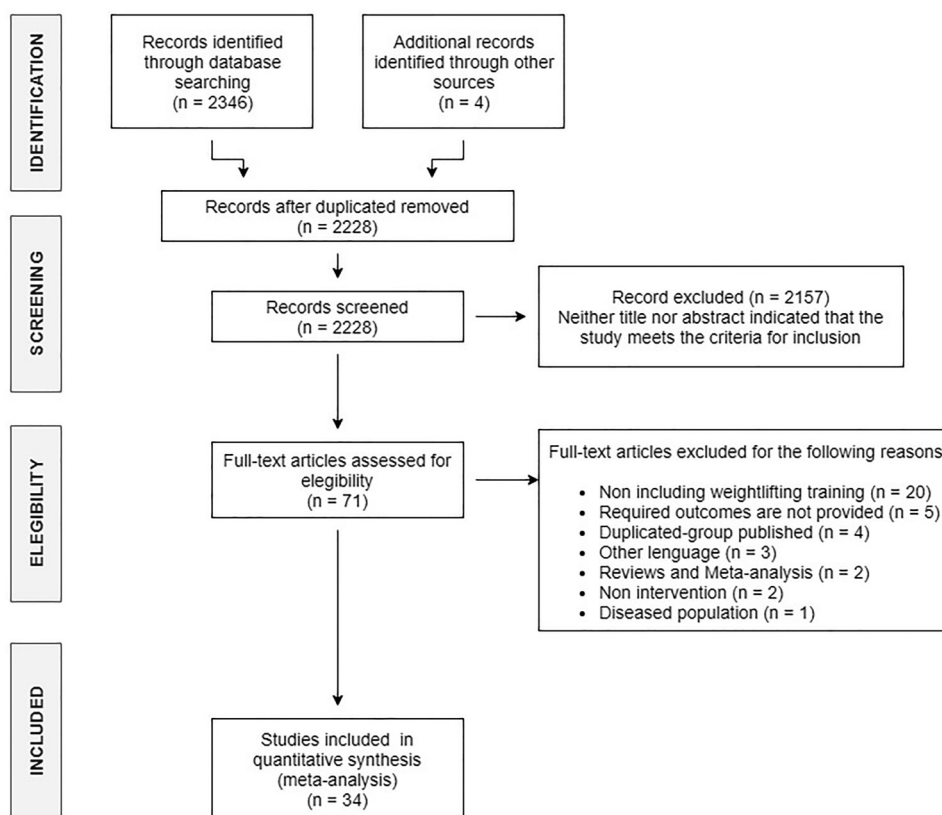


Figure 1. Selection of studies included in the meta-analysis. PRISMA flowchart.

method was used for imputing missed ESs. Three ESs were imputed for SJ (Figure 6(a)), six for CMJ (Figure 6(b)), and nine for 1RM (Figure 6(c)). After imputing missed ESs, the ES magnitudes were diminished for SJ (from 0.31 [95% CI = 0.23, 0.40] to 0.28 [95% CI = 0.18, 0.38]), for CMJ (from 0.33 [95% CI = 0.21, 0.46] to 0.26 [95% CI = 0.12, 0.41]) and for 1RM (from 0.58 [95% CI = 0.38, 0.79] to 0.42 [95% CI = 0.13, 0.71]). Therefore, publication bias cannot be disregarded as a possible threat to the validity of the current findings for SJ, CMJ, and 1RM.

Discussion

The aim of this meta-analysis was to analyze the effect of isolated weightlifting training, combined weightlifting and traditional resistance training on squat strength, jumping and sprinting performance improvements. Our analysis did not find a significant difference between training groups. However, weightlifting exercises combined with traditional resistance training exercises caused a greater magnitude of changes in CMJ, sprint, and 1RM.

The benefits of including weightlifting movements in training sessions compared to traditional resistance training to improve jumping performance could be dependent on the jump type. Thus, while TD showed a higher magnitude of changes on SJ performance, the groups including

weightlifting exercises (WL and CB) showed a greater magnitude of changes in CMJ height. Nevertheless, all groups showed a small effect in both jump types. A reason for the higher magnitude of change in CMJ when weightlifting exercises are used in training sessions could be attributed to changes in muscle-tendon stiffness,¹² which optimize the stretch-shortening cycle required in CMJ. Also, neuromuscular adaptations may explain the greater ES found in CMJ, since they have been linked to the ability to produce higher force levels during explosive movements after weightlifting training.^{22–26,71} Although there are differences between movements (e.g. weightlifting exercises involve bar acceleration, while the jump involves body mass acceleration), several authors have reported comparable values for peak force and rate of force development between weightlifting and jump tasks.^{22,25,72} In this sense, several authors^{22–26} have shown that weightlifting movement patterns are similar to vertical jump. So, the benefits on CMJ performance after WL and CB training found in our analysis may be related to changes in peak force and rate of force development.⁷³ Since plyometric exercise could increase the effect on vertical jump, the CB group effect could have been influenced by type training. So, a higher SMD was shown for training that combined weightlifting movements and plyometric exercise as in James' studies.^{57,58}

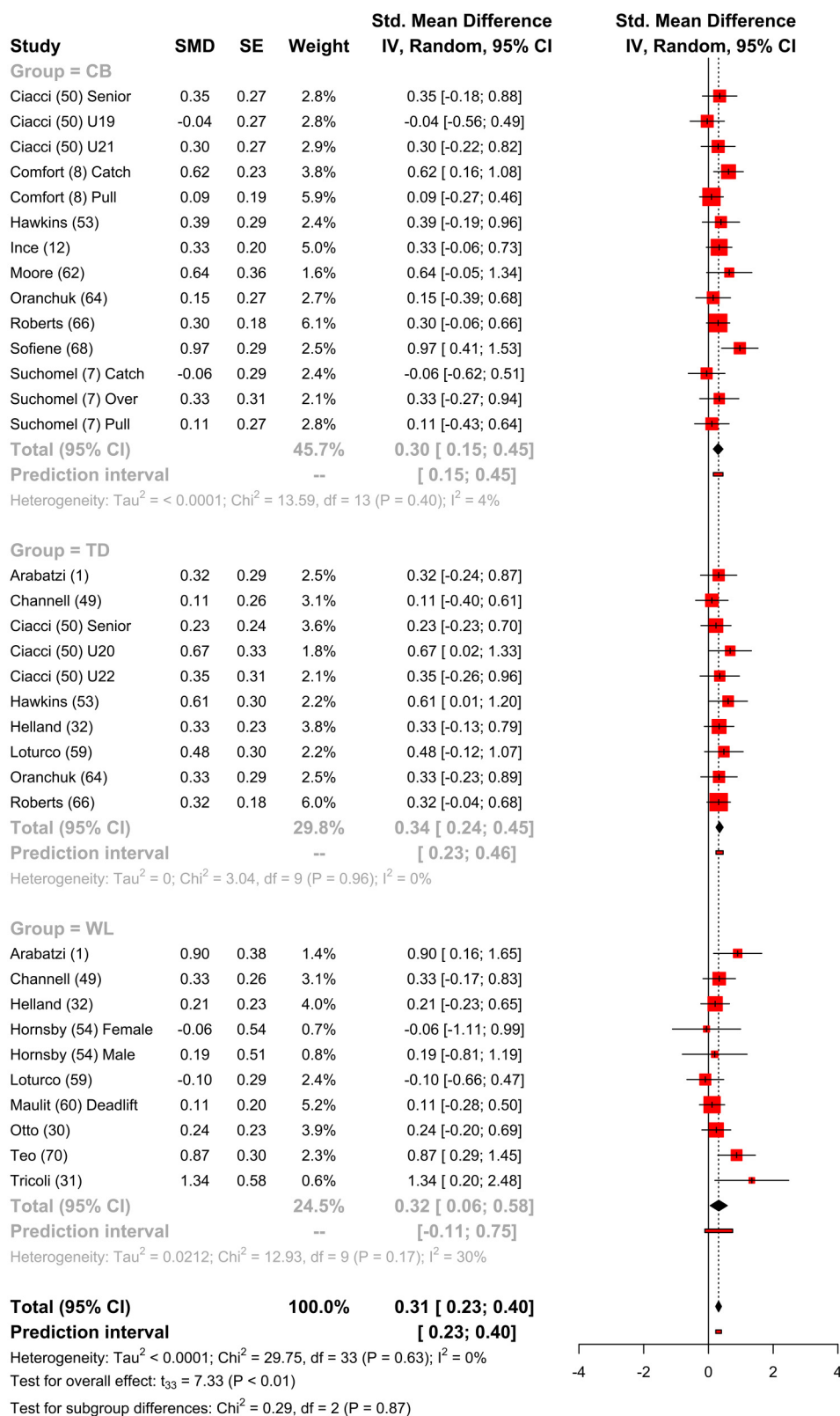


Figure 2. Meta-analysis of the effect of three type of exercises (CB = weightlifting and derivatives combine with traditional exercises; TD = only traditional exercises; WL = only weightlifting and derivatives exercises) on squat jump performance. Values on x-axis denote standardised mean differences (SMD). Random effects model with predictive interval. The predictive interval indicates the range within which we expect the effects of 95% of future studies will be. For studies that had multiple study group, the effect are presented independently and are marked as it is called on original paper.

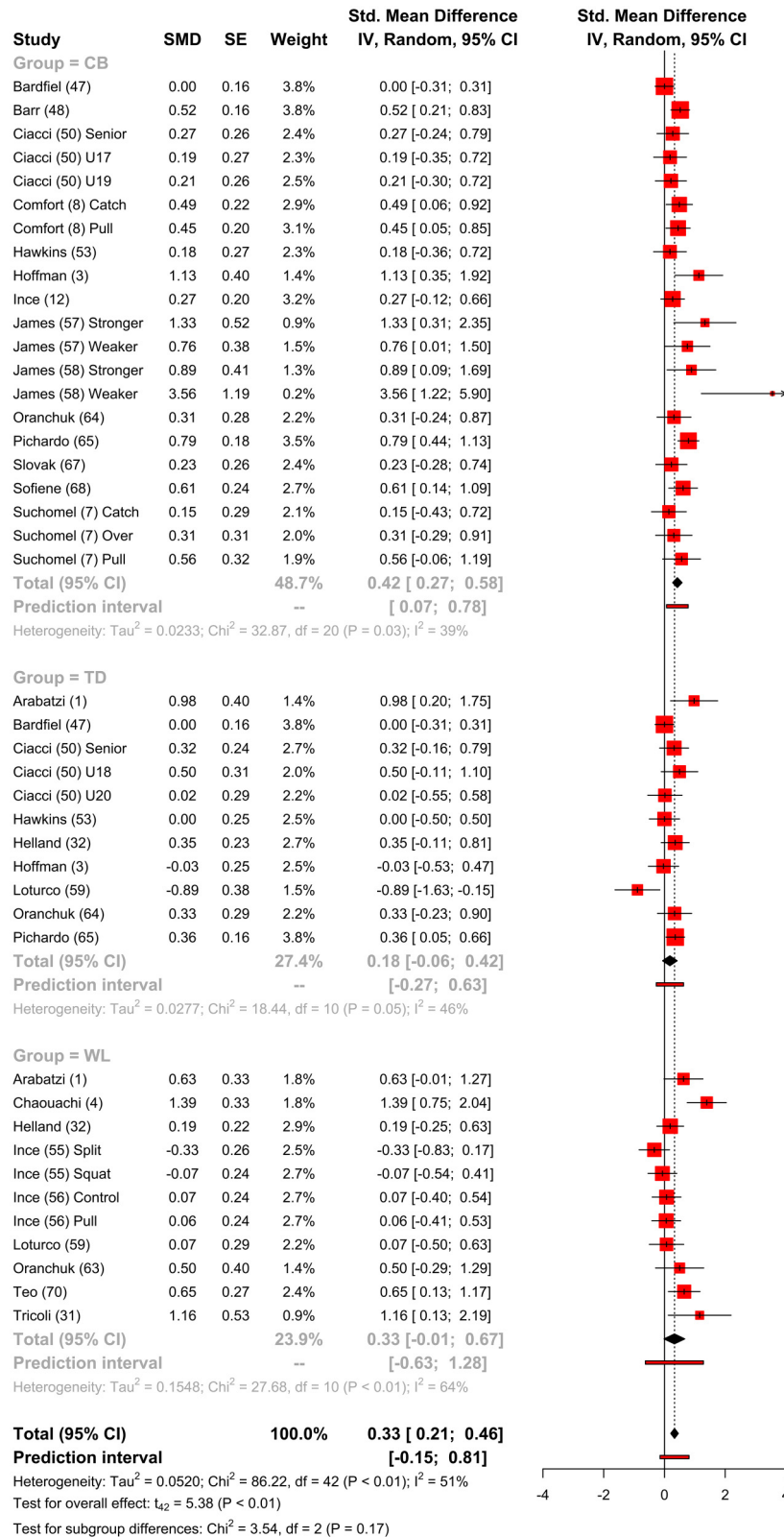


Figure 3. Meta-analysis of the effect of three type of exercises (CB = weightlifting and derivatives combine with traditional exercises; TD = only traditional exercises; WL = only weightlifting and derivatives exercises) on countermovement jump performance. Values on x-axis denote standarised mean differences (SMD). Random effects model with predictive interval. The predictive interval indicates the range within which we expect the effects of 95% of future studies will be. For studies that had multiple study group, the effect are presented independently and are marked as it is called on original paper.

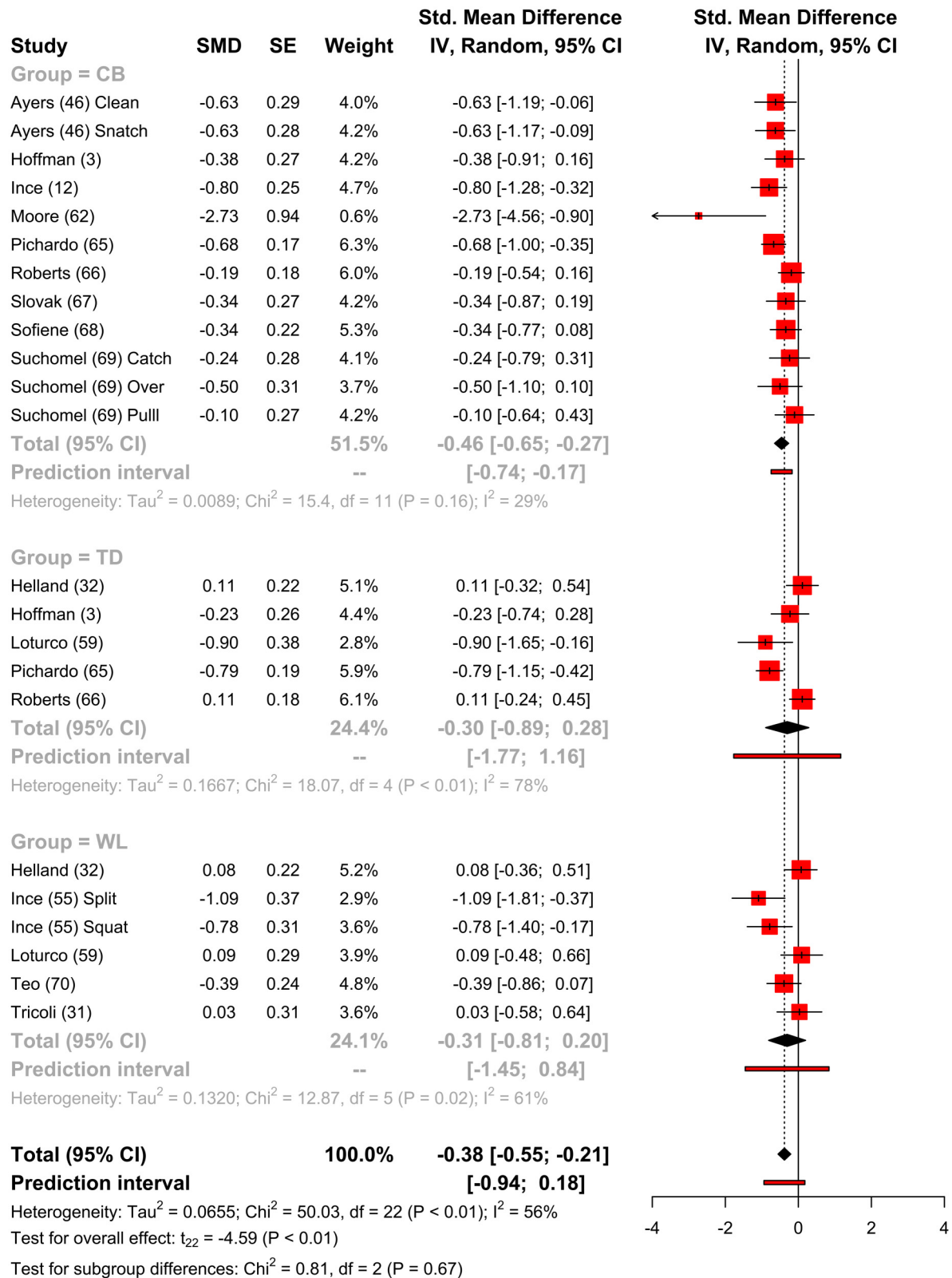


Figure 4. Meta-analysis of the effect of three type of exercises (CB = weightlifting and derivatives combine with traditional exercises; TD = only traditional exercises; WL = only weightlifting and derivatives exercises) on sprinting performance. Values on x-axis denote standardised mean differences (SMD). Random effects model with predictive interval. The predictive interval indicates the range within which we expect the effects of 95% of future studies will be. For studies that had multiple study group, the effect are presented independently and are marked as it is called on original paper.

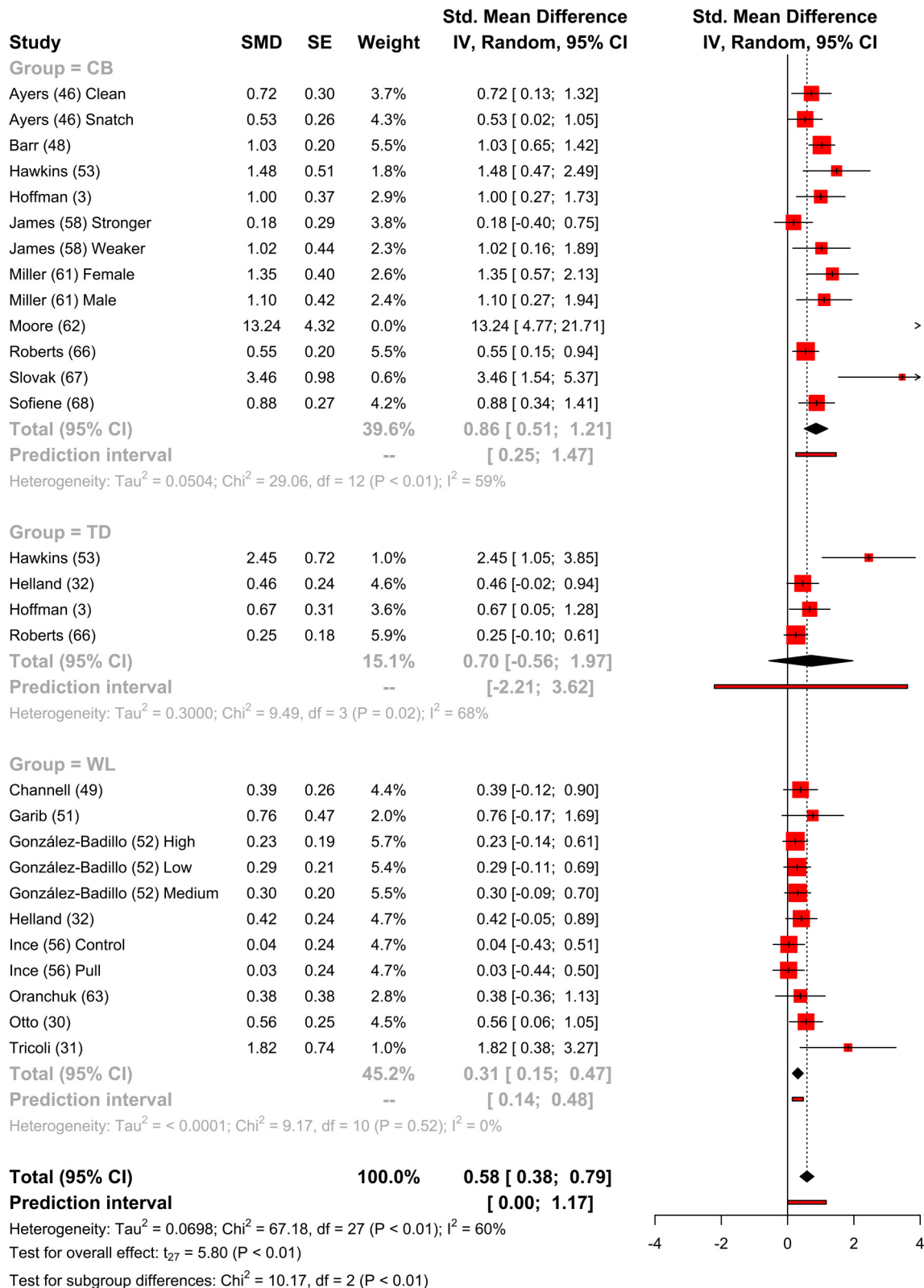


Figure 5. Meta-analysis of the effect of three type of exercises (CB = weightlifting and derivatives combine with traditional exercises; TD = only traditional exercises; WL = only weightlifting and derivatives exercises) on squat strength. Values on x-axis denote standardised mean differences (SMD). Random effects model with predictive interval. The predictive interval indicates the range within which we expect the effects of 95% of future studies will be. For studies that had multiple study group, the effect are presented independently and are marked as it is called on original paper.

Table 2. Results of analysing the influence of categorical moderators variables on effect for CMJ.

Moderator	Category	k	SMD [95% CI]	Q	p	I ²
Experience	Beginner	14	0.32 [0.02, 0.62]	5.63	0.228	50%
	Novice	13	0.36 [0.25, 0.46]			
	Intermediate	12	0.41 [0.15, 0.68]			
	Advance	2	0.77 [-2.26, 3.81]			
	Elite	2	-0.37 [-6.45, 5.69]			
Training days per week	≤ 2 days	24	0.21 [0.08, 0.33]	10.21	0.006	43%
	3 days	17	0.37 [0.37, 0.83]			
	≥ 4 days	2	0.51 [-6.86, 7.88]			
Weeks of intervention	≤ 8 weeks	18	0.22 [0.02, 0.41]	4.97	0.083	48%
	8 – 12 weeks	12	0.58 [0.28, 0.86]			
	> 12 weeks	13	0.30 [0.11, 0.48]			

Note. 95% CI: 95% Confidence Interval around SMD; Q: statistic for testing the significant of the moderator variable; I²: heterogeneity index; k: number of analysis' units; SMD: standard mean difference; p: probability level associated to Q.

Another important variable to interpret the results from studies can be the training load. There are differences in the total training load employed in weightlifting training to improve jump performance. For example, in Hawkins et al.'s⁵³ protocol participants have to complete repetition until muscle failure during training sessions. These authors showed benefits for SJ, but not for CMJ, so training with repetitions until failure may not be a good strategy to improve explosive movements that include the stretch-shortening cycle such as CMJ, but could be a possibility to improve concentric-only movements like SJ, as shown in the TD group. Nevertheless, another variable was confirmed—frequency of training—which is a moderator for improving CMJ performance. In this sense, the present meta-analysis showed that higher frequency (≥ 4 days per week) in studies such as that by Hoffman et al.³ achieved a higher effect compared with studies with lower training days per week, such as that by Ciacchi and Bartolomei⁵⁰ (≤ 2 days per week).

Concerning sprinting time, results obtained from meta-analysis showed an unusual result in the Moore et al.'s⁶² study, with a very high SMD according to other

studies. We repeated the analysis without that study, and the new analysis showed a lower SMD for the CB group (ES = -0.48) and lower heterogeneity (I² = 34%).

Sprinting performance improvements may be related to changes in the stretch-shortening cycle pattern and increases in the muscle-tendon unit's stiffness, which has been reported after weightlifting training.¹ In this context, some authors have linked muscle-tendon unit stiffness to lower-limb performance,⁷⁴ which explains the higher leg stiffness values found in power athletes compared with endurance athletes or untrained populations.^{75–77} According to the result of moderator analysis, the benefits of weightlifting on sprinting performance are mediated by participants' level. For example, Pichardo et al.⁶⁵ included participants aged between 9 and 14 years old and showed a moderate increase in sprinting performance (ES = -0.68), while Helland et al.³² measured high-level athletes, showing a lower effect (ES = -0.08) after weightlifting training. Another explanation for heterogeneity between studies may be related to exercises included in the training program. In Loturco et al.'s⁵⁹ study, one group carried out

Table 3. Results of analyzing the influence of categorical moderators variables on effect for sprint.

Moderator	Category	k	SMD [95% CI]	Q	p	I ²
Experience	Beginner	9	-0.71 [-0.91, -0.52]	41.75	<0.001	56%
	Novice	4	0.01 [-0.23, 0.26]			
	Intermediate	7	-0.30 [-0.41, -0.19]			
	Advance	1	0.03 [-0.58, 0.64]			
	Elite	2	-0.38 [-6.68, 5.93]			
Training days per week	≤ 2 days	12	-0.50 [-0.77, -0.24]	5.95	0.051	56%
	3 days	7	-0.31 [-0.63, 0.01]			
	≥ 4 days	4	-0.12 [-0.45, 0.20]			
Weeks of intervention	≤ 8 weeks	15	-0.33 [-0.55, -0.11]	2.68	0.262	56%
	8 – 12 weeks	4	-0.34 [-1.16, 0.49]			
	> 12 weeks	4	-0.58 [-0.97, -0.19]			

Note. 95% CI: 95% Confidence Interval around SMD; Q: statistic for testing the significant of the moderator variable; I²: heterogeneity index; k: number of analysis' unit; SMD: standard mean difference; p: probability level associated to Q.

Table 4. Results of analyzing the influence of categorical moderators variables on effect for 1RM squat.

Moderator	Category	k	SMD [95% CI]	Q	p	I ²
Experience	Beginner	4	0.70 [-0.31, 1.72]	5.24	0.155	59%
	Novice	10	0.76 [0.37, 1.14]			
	Intermediate	12	0.38 [0.13, 0.63]			
	Advance	2	1.14 [-5.33, 7.62]			
	Elite	—	—			
Training days per week	≤ 2 days	8	0.40 [0.04, 0.77]	2.89	0.235	56%
	3 days	11	0.86 [0.36, 1.37]			
	≥ 4 days	9	0.49 [0.22, 0.75]			
Weeks of intervention	≤ 8 weeks	20	0.53 [0.29, 0.77]	7.58	0.022	57%
	8–12 weeks	5	0.48 [-0.32, 1.27]			
	>12 weeks	3	0.94 [0.47, 1.41]			

Note. 95% CI: 95% confidence interval around SMD; Q: statistic for testing the significant of the moderator variable; I²: heterogeneity index; k: number of analysis' units; SMD: standard mean difference; p: probability level associated to Q.

weightlifting through the push press exercise, and another group performed jump squat. Push press could not be a specific task compared to jump squat to improve sprinting performance, despite Comfort et al.⁷⁸ have shown some similarities between squat jump and push press executions. This might partially explain the greater effects found in the TD group compared with the CB and WL groups.

The last variable analyzed was the effect of weightlifting training on 1RM in the squat exercise. This variable initially showed higher values of SMD in the CB and TD groups, but with an important influence of two studies. The unusual 1RM changes reported by Moore et al.'s⁶² (from 70 to 212 kg) and Hawkins et al.'s⁵³ (from 112.25 to 147.75) studies, made us contact the authors to ask for the original data. Unfortunately, no responses were obtained. For these reasons, we decided to repeat the

analysis for this variable without data from those studies.

As a result, SMD showed a moderate effect for CB (ES = 0.78) and small effect for TD (ES = 0.39).

The moderate effect on maximal strength after weightlifting training could be explained by the high relative loads used in weightlifting exercises. Loads above 80% of 1RM are required to maximize power output in weightlifting exercises like the SN or the C&J.⁷⁹ Therefore, the concomitant factor of the load that maximized power output and a heavy load can be postulated as an optimal stimulus to increase 1RM strength. In this line, Schoenfeld et al.⁸⁰ suggested that heavy load training is more effective than lower loads for improving maximal strength. Also, resistance training performed at higher movement velocity is a greater stimulus for enhancing cross-sectional area, force, and power than training at lower movement velocities.⁸¹

Table 5. Results of meta-analysis after removing outliers.

Variable	Subgroup	k	SMD [95% CI]	Q	p	I ²
SJ		33	0.30 [0.22, 0.38]	1.28	0.528	0%
	CB	13	0.26 [0.14, 0.38]			
	WL	10	0.31 [0.06, 0.45]			
	TD	10	0.34 [0.24, 0.45]			
CMJ		39	0.33 [0.24, 0.43]	2.76	0.252	26%
	CB	20	0.41 [0.28, 0.55]			
	WL	9	0.25 [0.00, 0.49]			
Sprint	TD	10	0.22 [0.05, 0.40]	0.72	0.696	52%
		22	-0.37 [-0.52, -0.21]			
	CB	11	-0.44 [-0.60, -0.29]			
	WL	6	-0.31 [-0.81, 0.20]			
1RM squat	TD	5	-0.30 [-0.89, 0.28]	15.41	<0.001	43%
		25	0.53 [0.38, 0.69]			
	CB	11	0.81 [0.57, 1.04]			
	WL	11	0.31 [0.15, 0.47]			
	TD	3	0.39 [-0.10, 0.87]			

Note. 95% CI: 95% confidence interval around SMD; Q: statistic for testing the significant of the subgroup; I²: heterogeneity index; k: number of analysis' units; SMD: standard mean difference; p: probability level associated to Q.

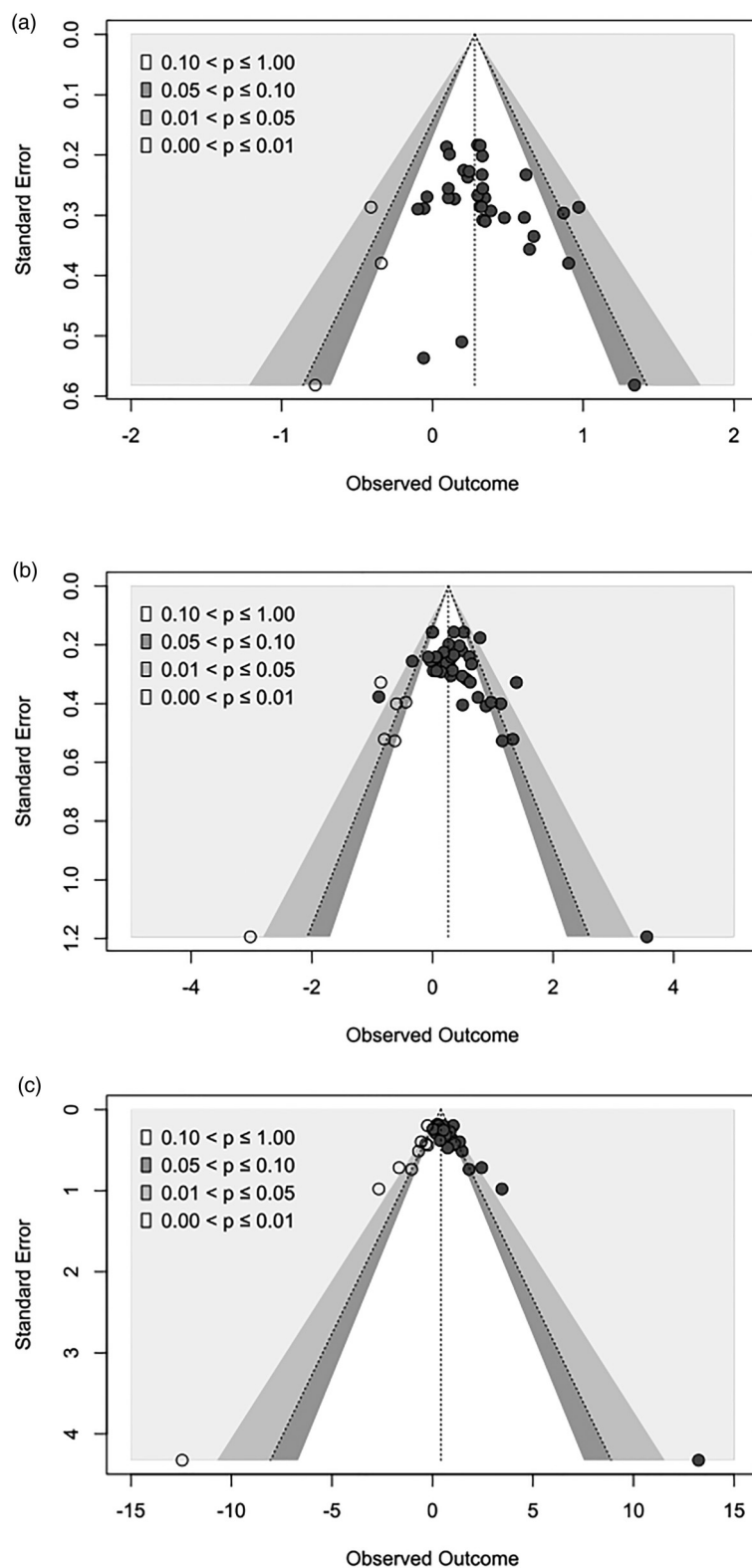


Figure 6. Analysis of publication bias for SJ (a), CMJ (b), and IRM squat (c). Results of trim-and-fill method are drawn (empty dot are imputed missed effect size). Funnel plot, using data from studies included in meta-analysis.

In this sense, weightlifting exercises require movements to be performed at maximal intended velocity and usually show higher velocities than traditional resistance exercises such as squat.⁸² Velocity is a key point during weightlifting performance and could be an important advantage to improve maximal strength through these movements. A final variable that could take in account, is the training program duration, where longer studies have shown a bigger effect on 1RM than shorter studies.^{3,48} Despite the usefulness of training programs based on weightlifting exercises in improving explosive actions, the current meta-analysis presents some limitations. An important limitation is the high variability found in participants' characteristics, including a wide range of ages and, in particular, differences in resistance training experience. Further, the diversity in training program variables (i.e. intensity, volume, program duration) is an important limitation to obtain more solid conclusions. Altogether, these limitations highlight the necessity for more studies, which will allow the detection of any qualitative moderator variable.

Practical application

The main finding in this meta-analysis is that weightlifting and their derivatives are effective for improving sprinting, jumping performance, and squat strength, as long as they are combined with other traditional resistance exercises. Based on the magnitude of changes, it should be highlighted that the combination of weightlifting with traditional resistance training appears to be the most useful strategy to improve 1RM squat, jumping and sprinting performance. Based on the present results, strength and conditioning coaches are encouraged to prescribe combined weightlifting training from 2 to 4 days per week for 8–12 weeks. According to prescription characteristics from studies on this meta-analysis, we can suggest that weightlifting exercises would be carried out over two to six series, with an intensity higher than 75% of 1RM. The use of specific exercises and intensities depending on the periodization phase remains understudied, although research is showing promising results.⁸³ In this sense, derivatives such as clean/snatch pull from floor, thigh, or knee are strongly recommended for several training phases.

Author contribution

The screening and writing of the manuscript were carried out by R.S., J.L., A.M. and A.G.; the data analysis was performed by A.G. with support from A.M.; the methodology was supervised by J.L. and A.M.; and, the original idea was conceived by R.S. who supervised the project.

Declaration of conflicting interests

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